

# Applying Artificial Intelligence to Cardiac MRI to Diagnose Congenital Heart Disease in Low-Resource Settings such as Sub-Saharan Africa

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This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Thank you for your excellent manuscript. Here are my comments.

1. Although this is a narrative review, your manuscript would benefit from greater transparency regarding the methodology for literature selection. Please clarify whether a structured search strategy was used (e.g., databases searched, inclusion/exclusion criteria, search terms). Even if not systematic, a brief methodological paragraph would enhance reproducibility and reduce selection bias concerns.

2. Your many cited examples (e.g., Nigeria, Ethiopia, South Africa) focus on relatively higher-capacity SSA countries. Please discuss the extent to which the recommendations can be generalized across SSA. Are there distinct regional or linguistic barriers that should be acknowledged? Consider adding a table or figure summarizing national-level CMR capacity if data permit.

3. While your manuscript extensively discusses systemic barriers, there is limited explicit discussion of confounding factors—especially inter-study variability and heterogeneity in healthcare access, provider workflows, or AI model performance. Please consider adding a section briefly reflecting on unmeasured and residual confounding, such as the influence of informal health systems, rural/urban disparities, or international equipment donation patterns.

4. You acknowledge the limited availability of SSA-derived imaging datasets but do not fully explore how this impacts AI model generalizability. Please strengthen this point by referencing specific examples of AI model failures or limitations due to domain shift when applied outside of their training population (e.g., ethnic bias in segmentation, site-specific imaging protocol variation).

5. While your limitations are embedded throughout, a clearly labeled section toward the end of the review is missing. Please add a standalone "Limitations" section summarizing key issues, including a) Lack of SSA-specific CMR datasets b) uneven infrastructure distribution c) Lack of economic evaluations of AI-enabled CMR and d) Regulatory uncertainty for AI deployment in SSA.

6. Your figures are excellent and informative, especially Figures 2–8. Please consider adding a summary table comparing diagnostic modalities (Echo vs CMR vs CT) in CHD across LMIC settings, especially in SSA.

7. Please define key AI terms (e.g., federated learning, compressed sensing) briefly on first use for our readers.

8. Your manuscript commendably addresses key ethical and medicolegal challenges of AI deployment (e.g., data privacy, liability concerns), which are often under-discussed in implementation reviews. However, please consider briefly suggesting a policy framework or model regulation to guide AI governance in LMICs (e.g., African Union Digital Health frameworks, WHO guidelines on AI ethics).

Reviewer #2

(Remarks to the Author)

Thank you for sharing this very well written manuscript.

The data underlying the manuscript is very well researched and the authors demonstrate deep knowledge of the subject being discussed.

A few very minor comments:

1. Line 291: "following" should rather be "followed by"

2. Line 190: You describe the need for manuals to be translated into local language. Can you clarify the target for these manuals? I imagine that technicians who will be involved in the care and maintenance of these equipment should be literate individuals with minimum level of technical expertise to avoid damage to these very expensive equipment.

3. Consider abbreviating the text in the opening paragraphs that describes the levels of evidence as some of these classifications are widely known.

4. Consider limiting your abbreviations to those that are not widely known.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Thank you for your rebuttal letter. I have no further comments.

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## **Point-by-Point Response to Reviewer Comments**

### **Reviewer #1**

#### **1. Literature Selection Methodology**

**Response:** Thank you for this insightful comment. A new “Methods” section was added early in the manuscript to clarify our literature selection strategy. While the review remains narrative, we now describe our flexible search across PubMed, Scopus, and Google Scholar using combinations of relevant terms. Inclusion criteria and synthesis methods are also outlined.

#### **2. Generalizability Across SSA and Regional Barriers**

**Response:** We expanded the section on infrastructure disparities to emphasize regional variation across SSA. We now discuss the implications of linguistic diversity, regional instability, and uneven infrastructure on generalizability. References were added, and we note that while the highlighted examples are from higher-capacity countries, similar challenges exist throughout the region.

#### **3. Unmeasured Confounding and Inter-Study Variability**

**Response:** We added a dedicated subsection under “Data Limitations” titled “Systemic and Contextual Variability in SSA,” which discusses confounders like informal healthcare systems, donor-driven equipment flows, urban-rural disparities, and inconsistent protocols. These confounders are acknowledged as important limitations when interpreting implementation results.

#### **4. AI Generalizability and Domain Shift**

**Response:** The section on “Bias in AI Models” now includes expanded discussion on domain shift, citing examples from retinal imaging and chest radiography. We added references to illustrate how ethnic and protocol variation can degrade model performance and emphasized the need for local retraining of CMR algorithms in SSA.

#### **5. Standalone “Limitations” Section**

**Response:** A new, clearly labeled “Limitations” section has been added near the end of the manuscript summarizing the five key issues raised: (1) lack of SSA-specific CMR datasets, (2) uneven infrastructure distribution, (3) absence of cost-effectiveness analyses, (4) regulatory uncertainty, and (5) narrative review design limitations.

#### **6. Diagnostic Modality Comparison Table**

**Response:** Table 2 was added comparing echocardiography, CMR, and CT in the SSA context, specifically for CHD evaluation. It includes strengths, limitations, radiation exposure, and availability status in SSA. This complements the discussion and supports implementation decisions.

#### **7. Definition of Key AI Terms**

**Response:** We now define terms such as “federated learning” and “compressed sensing” at first mention in the “Role of AI” section. These definitions aim to improve clarity for general readers unfamiliar with advanced AI concepts.

#### **8. Policy Frameworks for AI Governance in SSA**

**Response:** In the section on “Ethical and Legal Concerns,” we incorporated discussion of the

African Union’s Digital Transformation Strategy and WHO’s AI Ethics Guidance. These frameworks are briefly summarized as potential models for future SSA-specific AI regulation.

**Reviewer #2**

1. **Line 291: Language Correction**

**Response:** “Following” has been corrected to “followed by” as suggested.

2. **Line 190: Clarification on Manual Translation Target**

**Response:** We revised this sentence to clarify that the translation of maintenance manuals targets locally trained technicians or mid-level staff, who may lack fluency in technical English. Formal biomedical engineers are typically proficient in English or French, but practical operation is often delegated.

3. **Opening Paragraphs on Evidence Levels**

**Response:** We abbreviated the paragraph describing levels of evidence and focused on the key guideline points relevant to CMR’s role in CHD, as per your suggestion.

4. **Abbreviation Use**

**Response:** We revised the manuscript to limit abbreviations to less commonly known terms. For those that remain, we have provided a list of acronyms at the beginning to avoid overcrowding the tables and preserve readability.

Sincerely,

On behalf of all co-authors,

Michael Negussie, MD  
School of Medicine  
Addis Ababa University  
Addis Ababa, Ethiopia